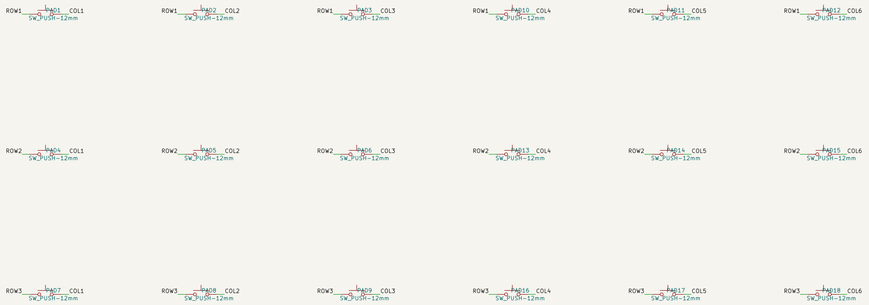
DECK-PAD Bukifon v1.0 290 x 78 x 1.6 mm (spec/10-panel-ui.md 3.1, spec/20-electronics.md 6.3) CERN-OHL-S-2.0

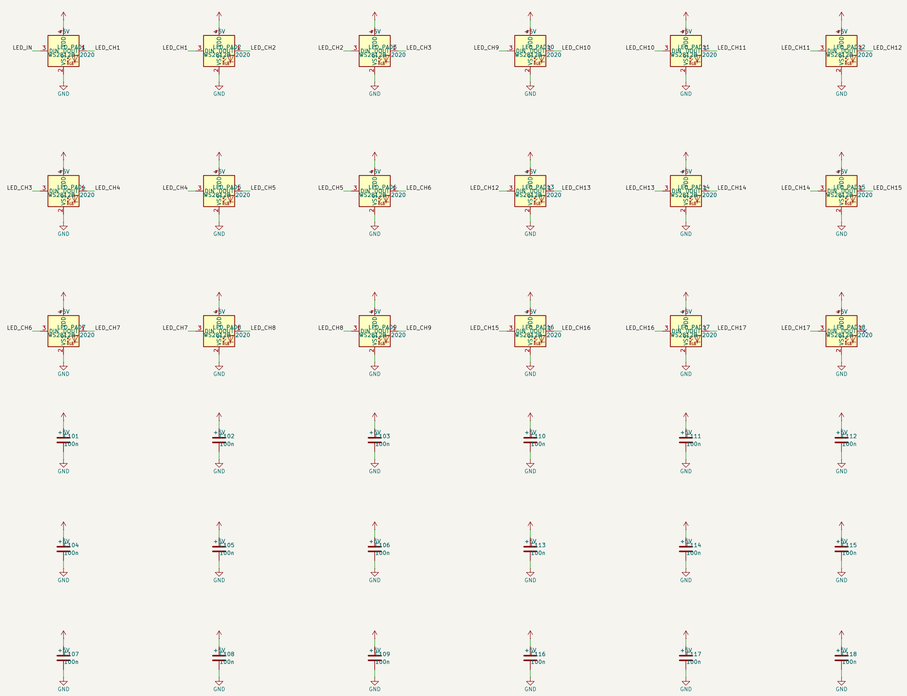
18 tactile 12x12x5 + 18x WS2812B-2020 + 2x 30 mm fader (10k, 3V3) + MCP23017 (0x20, matrix 3x6) + 2x 74HC4067 (FSR option) + IDC-16

PAD1..PAD18 matrix 3 rows x 6 columns (ROW1..3 = MCP23017 GPA0..GPA2 driven low, COL1..6 = GPB0..GPB5 read with external 10k pull-up R21..R26). Panel coordinates in matrix-tables.md.

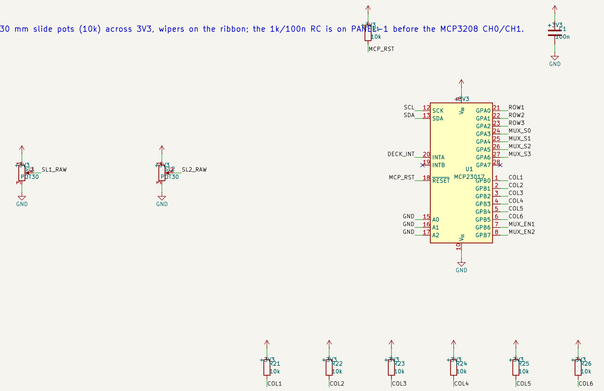


FSR option: 18 x 2-pin pad (film lead) + 10k pull-up to 3V3, one node per mux channel; channel select S0..S3 = MCP23017 GPA3..GPA6, enable = GPB6/GPB7. When the FSR film is not fitted the channel reads full scale (3V3) = normal.

18 x WS2812B-2020 in one chain after LED_IN from PANEL-1, powered from +5V (J-DECK pin 16), 100n per LED. Last DOUT = no connect. Brightness is limited by firmware (total 9+18 or 9+42 LEDs).



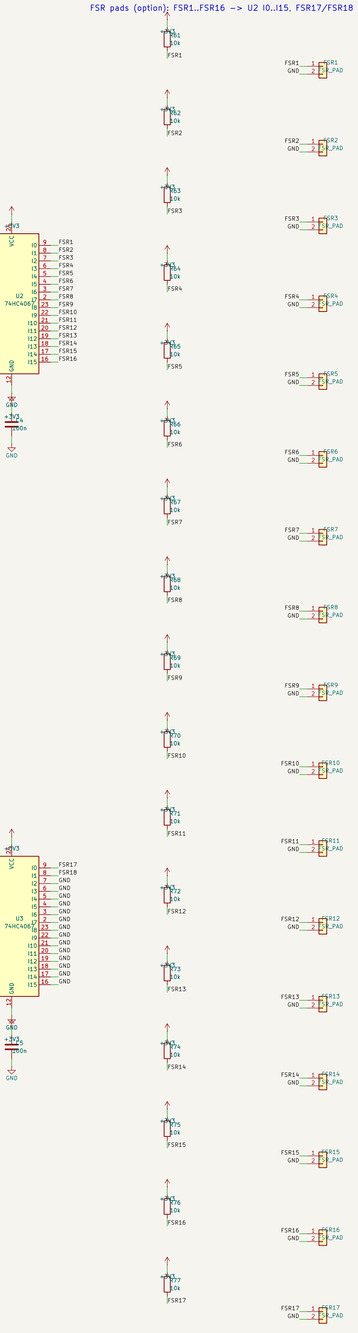
SL1/SL2 = 30 mm slide pots (10k) across 3V3, wipers on the ribbon; the 1k/100n RC is on PANEL-1 before the MCP3208 CH0/CH1.

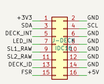


DECK_ID for DECK-PAD: 33k / 3.3k -> 3.3*3.3/36.3 = 0.30 V (spec/20-electronics.md 6.2: 0.30 V +-0.15 V)



FSR pads (option): FSR1..FSR16 -> U2 I0..I15, FSR17/FSR18 -> U3 I0/I1.





Bukifon open hardware — CERN-OHL-S-2.0

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File: deck-pad.kicad_sch

Title: Bukifon DECK-PAD — 290x78 — 18 pads 12x12 / 18xWS2812B / 2 fa

Size: A1	Date:	Rev: 1.0
KiCad E.D.A. 10.0.4		Id: 1/1